

# ARYA WIDURA SATRIA WIBOWO HARYANTO

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A highly motivated Electrical Engineering graduate with a solid foundation in control systems, process instrumentation, and industrial automation. Interested in applying engineering principles to enhance reliability, efficiency, and safety across complex operations. Eager to contribute in a safety-critical industrial environment, supported by strong analytical thinking, adaptability, effective communication, and a continuous desire to grow professionally.

## Work Experiences

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### PT Medco E&P Indonesia - Jakarta(On Site)

Jul 2024 - Aug 2024

#### Junior Instrument Engineer Internship

- Reviewed 15+ instrument datasheets and prepared Material Take-Off (MTO) documents for over 30 instrumentation items, ensuring technical accuracy for procurement and installation in accordance with engineering standards
- Utilized AutoCAD to read, modify, and produce technical drawings such as P&IDs, hook-up diagrams, and instrument layouts to support field implementation
- Participated in alarm management review meetings, gaining insight into alarm prioritization, nuisance alarm reduction, and operator response strategies aligned with ISA-18.2
- Verified 10+ shutdown and interlock logic signal paths by validating 4–20 mA analog signals between transmitters and final control elements using EcoStruxure Control Expert, in line with loop check and pre-commissioning procedures based on ISA-5.1

### PT TJB Power Services - Jepara(On Site)

Jan 2024 - Feb 2024

#### Junior Instrument Engineer Internship

- Gained knowledge of steam power plant operations, including process flow and control systems, as well as Variable Frequency Drive (VFD) configuration on coal feeder systems to optimize feed control and its integration with DCS for automated process control
- Observed and learned calibration, verification, and documentation processes for 15+ field instruments, including level transmitters, pressure transmitters, and control valves
- Observed and learned loop checking procedures and basic troubleshooting steps on 10+ transmitters during field activities

### Industrial Automation Laboratory - Purbalingga(On Site)

Aug 2024 - Dec 2024

#### Assistant Coordinator

- Coordinated industrial automation practicum with 5 assistants and 30+ participants
- Collaborated with assistants to develop structured, relevant learning materials for the upcoming 6-month practicum
- Also contributed to troubleshooting logic errors and guiding real-time simulations, reinforcing practical understanding of PLC operations
- Delivered technical assistance in PLC-based traffic light simulation project using STEP 7 and Factory I/O, achieving 90% student completion rate

### Control System Laboratory - Purbalingga(On Site)

Aug 2024 - Dec 2024

#### Assistant

- Assisted in delivering control system practicum sessions involving PID tuning, feedback control simulations, and system response analysis using MATLAB/Simulink
- Guided 20+ students in understanding open-loop and closed-loop behavior, control system stability, and real-time simulation of dynamic systems
- Contributed to developing lab worksheets and exercises aligned with course objectives and industrial control standards

## Education Level

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### Jenderal Soedirman University - Purbalingga

Aug 2021 - May 2025

#### Bachelor of Electrical Engineering, 3.79/4.00

- Honors : Cumlaude
- Relevant Coursework : Instrumentation and Control Systems, Industrial Automation, Process Control, Electrical Measurements, Electrical System Design, PLC Programming, HMI Design, Embedded Systems, Digital Electronics, Microcontroller Applications
- Thesis : Design of Control and Safety System in Production Separator Based on PLC and HMI(2025)
- Actively participated in over 10 campus events, including technical webinars, seminars, and skill development workshops.

## Organisational Experience

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### Electrical Engineering Student Association - Purbalingga

Jan 2024 - Dec 2024

#### PLC Community Leader

- Led a community of 200+ members in the field of PLC and industrial automation, including developing a training curriculum for 1 year
- Organized 10+ technical workshops with avg. 40+ participants, covering topics like PLC programming and industrial sensors

- Delivered practical training on industrial automation using Siemens PLC to 20+ participants, effectively bridging theoretical knowledge with industry-relevant applications

## Skills, Project & Certification

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- **Hard Skills:** Instrumentation & Control System Design, Loop Checking, Instrument Calibration, Electrical Wiring, P&ID Interpretation, Troubleshooting, IoT System Design, Sensor Integration, Data Communication, Database
- **Tools:** TIA Portal, STEP 7, EcoStruxure Control Expert, WinCC, Vijeo Designer, AutoCAD, Arduino IDE, Factory I/O, MySQL, Firebase, Microsoft Office
- **Soft Skills:** Analytical Thinking, Problem Solving, Attention to Detail, Adaptability, Leadership, Team Collaboration, Effective Communication, Eagerness to Learn, Time Management
- **Projects:** 4-Way Traffic Light System with Omron PLC(2023), IoT-Enabled Medicine Box with Android and Firebase(2024)
- **Certification:** Basic Occupational Safety and Health (K3) - Realtech Academy(2024)